

Mirror Observerhood Lab V

Recursive Reliability Under Estimator Corruption

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Mirror Programme, Volume I: Observerhood — Computational Observerhood Labs

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Abstract

The first four Observerhood Labs tested when first-order reliability becomes viability-relevant. Lab V asks what happens when the reliability estimate itself becomes unreliable. A minimal viability-constrained decision process compares four agents: a no-reliability baseline, a first-order reliability agent, a recursive reliability agent and an oracle upper bound. Across 39,200 deterministic episodes, recursive reliability outperforms first-order reliability under reliability blindness, estimator corruption, false-alarm reliability and noisy residuals, but underperforms when first-order reliability is already truthful or when meta-diagnosis is too costly. The result supports a conditional version of recursive observerhood: meta-reliability is useful when lower-order reliability can fail in viability-relevant ways and when the cost of detecting that failure is lower than the avoided loss.

Keywords: Mirror Programme; Observerhood Labs; observerhood; recursive observerhood; self-models; self-model reliability; recursive reliability; meta-reliability; estimator corruption; reliability blindness; false alarms; viability; perturbation; computational cognition; symbolic artificial intelligence; constraint systems.

Contents

1	Roadmap position	3
2	Core claim	3
3	Scope and negative commitments	3
4	Formal framing	4
5	Experimental design	4
5.1	Decision process	4
5.2	Agent architectures	4

5.3	Perturbation regimes	4
5.4	Episodes and reproducibility	5
6	Results	5
7	Interpretation	7
8	Relationship to the Mathematics Arc	7
9	Limitations	8
10	Reproducibility statement	8
11	Conclusion	8

1 Roadmap position

This release forms part of the Computational Observerhood Labs of Mirror Programme, Volume I: Observerhood. The programme-level overview introduces a shared spine of constraint, model, reliability, repair, memory, viability, continuity and scale [1]. The Volume I overview applies that spine to observerhood as a derived organisational condition rather than a primitive subject or witness [2]. Mirror Theory I derives recursive observerhood from bounded viability-constrained organisation, including compressed self-models and reliability estimates [3]. Mirror Mathematics I supplies the first formal calculus for constraint regimes, local modelling, viability and reliability salience [4].

Lab I showed that scalar self-model reliability can improve viability under false self-location [5]. Lab II showed that decomposed reliability is informative but insufficient unless diagnosis is coupled to effective repair [6]. Lab III showed that reliability becomes viability-relevant only when it is actionable and cost-sensitive [7]. Lab IV then showed that reliability-gated repair has a bounded positive threshold region rather than universal advantage [8]. Lab V completes this first reliability sequence by testing recursion itself: when does an agent need to estimate whether its own reliability estimate should be trusted?

2 Core claim

The core claim of Lab V is conditional:

Recursive reliability improves viability where first-order reliability can itself fail in an action-relevant way, and where the expected benefit of detecting that failure exceeds the cost of meta-diagnosis.

The negative side is equally important. Recursion is not a free good. If first-order reliability is already truthful, meta-monitoring adds cost without benefit. If meta-diagnosis is too expensive, recursive reliability can become worse than trusting a simpler first-order mechanism.

3 Scope and negative commitments

This study is a minimal computational stress test, not a biological or phenomenological model of consciousness. The hidden state is binary, the reliability estimates are simulated rather than learned and the recursive monitor is hand-coded. The purpose is not to show that recursion is sufficient for observerhood, but to identify a narrow viability condition under which second-order reliability becomes functionally useful.

The paper makes five negative commitments:

1. It does not claim that recursive reliability is always beneficial.
2. It does not claim that deeper recursion should continue indefinitely.
3. It does not claim that a binary hidden-state task captures biological selfhood.
4. It does not treat the oracle as a realistic agent, but only as an upper-bound reference.
5. It does not rely on unpublished mathematics papers as prior authority; it supplies computational scaffolding for the later threshold synthesis.

4 Formal framing

Let M_L denote a local self-model and let $R_L^{(1)}$ estimate whether that self-model is currently accurate. A recursive reliability variable $R_L^{(2)}$ estimates whether the first-order reliability estimate is itself actionably trustworthy. The recursive salience expression is

$$\Sigma_L(R_L^{(2)}) = \mathbb{E}[V_L \mid M_L, R_L^{(1)}, R_L^{(2)}] - \mathbb{E}[V_L \mid M_L, R_L^{(1)}] - K_L(R_L^{(2)}),$$

where $K_L(R_L^{(2)})$ is the cost of maintaining, querying and acting on the recursive monitor.

Prediction 1 (Recursive reliability). *A recursive reliability agent should outperform a first-order reliability agent when the first-order reliability estimate is systematically corrupted, provided that meta-diagnostic cost is lower than the expected viability loss from trusting the corrupted estimate.*

5 Experimental design

5.1 Decision process

The environment is a minimal viability-constrained decision process. There is a hidden self-relevant state z and an internal self-model m . Acting while $m = z$ produces viability gain. Acting while $m \neq z$ causes viability loss. A first-order reliability variable $R^{(1)}$ estimates whether the self-model is accurate. In some conditions, that estimate becomes blind, false-alarming or corrupted.

The recursive agent uses a second-order consistency signal, here called a residual, to decide whether $R^{(1)}$ should be trusted, repaired or ignored. Residuals are fallible: they can be noisy, and querying them imposes meta-cost. This keeps recursion cost-bounded rather than unlimited.

5.2 Agent architectures

Four agents are compared:

1. **No-reliability baseline.** Acts from the self-model without reliability repair.
2. **First-order reliability agent.** Repairs the self-model when $R^{(1)}$ falls below threshold.
3. **Recursive reliability agent.** Audits conflicts between $R^{(1)}$ and residual evidence before deciding whether to repair or suppress false alarms.
4. **Oracle upper bound.** Repairs whenever the hidden self-state is wrong. This is not a deployable agent; it is an upper-bound comparison.

5.3 Perturbation regimes

Seven conditions are tested: control, truthful self perturbation, reliability blindness, estimator corruption, false-alarm reliability, noisy residual and high meta-cost. These conditions separate cases in which first-order reliability is already adequate from cases in which first-order reliability fails in a viability-relevant way.

5.4 Episodes and reproducibility

The experiment uses 1,400 episodes per agent-condition pair. Across seven conditions and four agents, this produces 39,200 deterministic episodes. The accompanying reproducibility package contains the single-file Python implementation, fixed-seed data, summary tables, recursive advantage table and figure-generation code.

6 Results

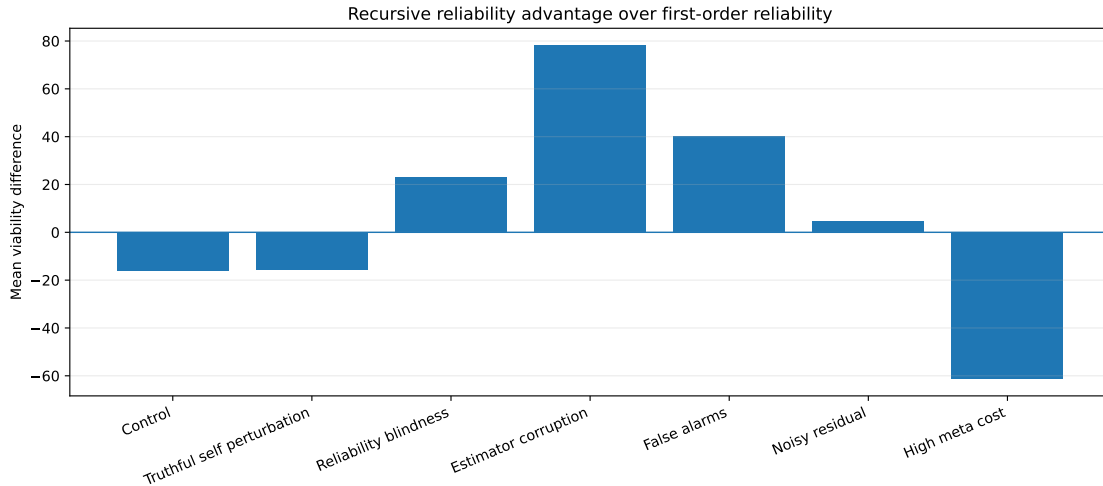


Figure 1: Recursive reliability advantage over first-order reliability. Positive values indicate regimes where auditing first-order reliability improves viability; negative values indicate unnecessary recursion or excessive meta-cost.

Table 1: Mean viability advantages of the recursive reliability agent. Values are means across 1,400 episodes per agent-condition pair.

Condition	Recursive – first-order	Recursive – no reliability	Oracle gap
Control	-16.0	-16.0	16.0
Truthful self perturbation	-15.7	185.5	15.6
Reliability blindness	23.0	185.7	19.3
Estimator corruption	78.3	183.6	21.6
False alarms	40.1	78.8	36.0
Noisy residual	4.8	145.9	40.8
High meta cost	-61.4	42.7	162.8

The strongest positive result appears under estimator corruption, where recursive reliability exceeds first-order reliability by 78.3 viability points. It also improves under false-alarm reliability by 40.1 points, reliability blindness by 23.0 points and noisy residuals by 4.8 points. These are precisely the regimes in which the first-order reliability estimate can itself become misleading.

The negative cases are equally informative. Under control, recursive reliability loses 16.0 points because there is nothing for the meta-monitor to correct. Under truthful self perturbation,

first-order reliability is already adequate and recursion again loses 15.7 points. Under high meta-cost, recursive reliability fails sharply, losing 61.4 points against first-order reliability because the cost of auditing reliability overwhelms the avoided loss.

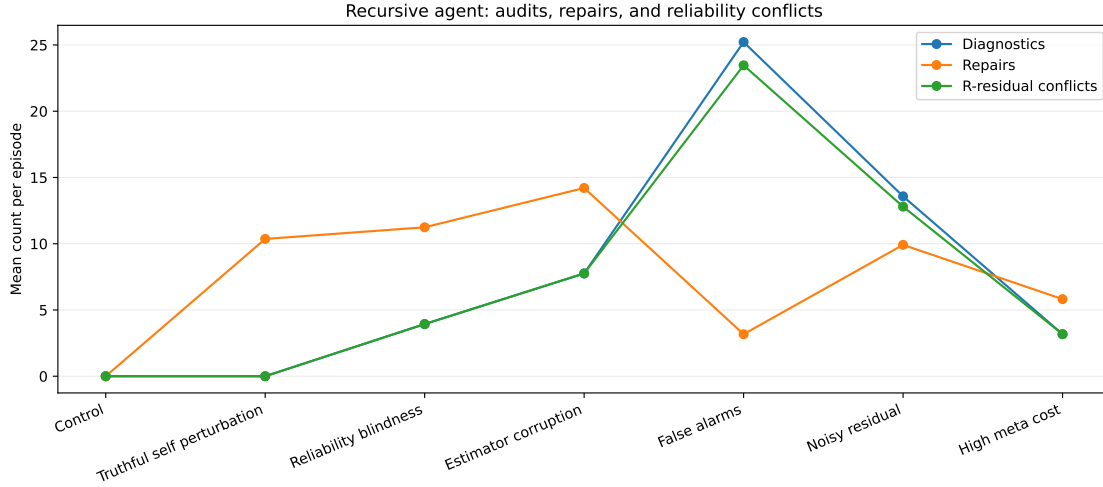


Figure 2: Recursive agent profile: diagnostics, repairs and reliability-residual conflicts by condition. The recursive monitor is most active where first-order reliability is vulnerable.

Table 2: Recursive agent diagnostic profile. Values are means per episode.

Condition	Repairs	Diagnostics	Conflicts	Wrong acts
Control	0.0	0.0	0.0	0.0
Truthful self perturbation	10.4	0.0	0.0	0.0
Reliability blindness	11.2	3.9	3.9	0.0
Estimator corruption	14.2	7.8	7.8	0.0
False alarms	3.2	25.2	23.5	0.0
Noisy residual	9.9	13.6	12.8	1.0
High meta cost	5.8	3.2	3.2	0.0

The profile shows why the result should not be read as a simple preference for recursion. The recursive agent performs no diagnostics under control or truthful self perturbation, yet still pays the standing meta-cost. It becomes useful where conflicts between first-order reliability and residual evidence identify reliability blindness, estimator corruption or false alarms. Under high meta-cost, the same mechanism becomes maladaptive despite detecting the right kind of failure.

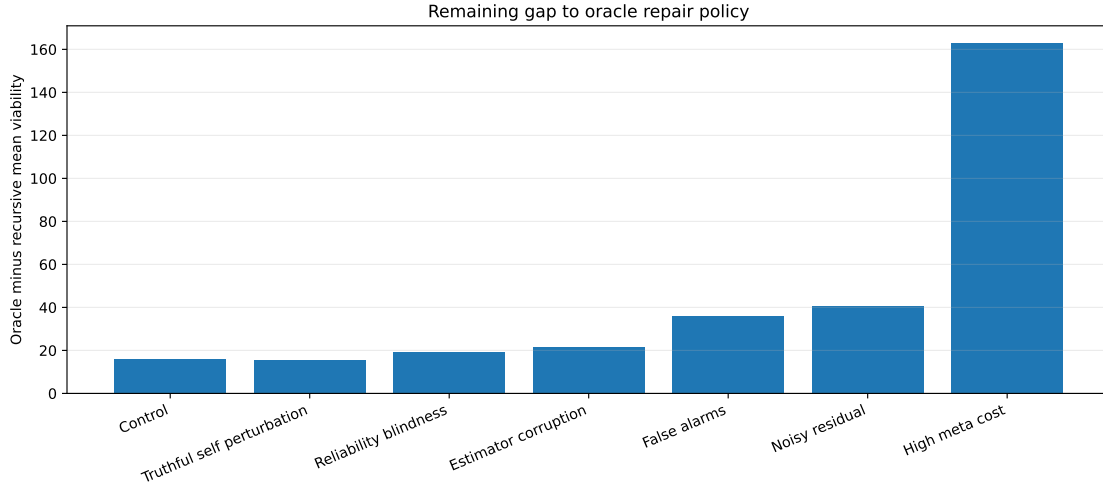


Figure 3: Remaining gap to the oracle repair policy. The oracle is an upper bound, not a deployable agent. The gap remains large where meta-diagnosis is costly or residual evidence is noisy.

The three numbered figures in the main paper focus on the recursive reliability claim: advantage over first-order reliability, the recursive agent profile and the remaining oracle gap. The reproducibility package also includes generated mean-viability, survival-rate and self-error figures. Those additional figures are consistency outputs rather than separately numbered main-text figures.

7 Interpretation

Lab V separates three levels: self-modelling, estimating whether the self-model is reliable and estimating whether that reliability estimate is itself trustworthy. The third level is useful only when the second level can fail in a viability-relevant way.

The result also disciplines the phrase “recursive observerhood”. It should not mean infinite self-reference. In the Mirror framework, recursive reliability is useful only as cost-bounded diagnosis of reliability failure. The recursion stops where additional monitoring no longer improves expected viability after cost.

This is why the high-meta-cost condition matters. It shows that even correct higher-order diagnosis can become non-salient if the act of diagnosing reliability failure costs more than the avoided loss. Recursive reliability is therefore conditional, not magical.

8 Relationship to the Mathematics Arc

Lab V completes the first reliability arc of the Computational Observerhood Labs. Labs I through IV establish that self-model reliability can be viability-positive, that decomposition alone is insufficient, that reliability must be actionable and cost-sensitive and that positive salience has a threshold structure. Lab V adds the recursive case: reliability over reliability becomes useful only when first-order reliability itself can fail in a costly, action-relevant way.

This supplies the final computational scaffold needed before Mirror Mathematics II. The later

formal paper can now cite Labs I through V as the experimental sequence motivating minimal observer thresholds: first-order reliability, channel decomposition, actionability, threshold salience and recursive reliability.

9 Limitations

The limitations are substantive.

First, the hidden state is binary. This makes failure modes interpretable, but it does not capture rich biological or artificial self-models.

Second, residual evidence is simplified. Real agents would need learned, noisy and context-sensitive diagnostics of reliability failure.

Third, the recursive policy is hand-coded. The experiment tests whether such a policy can be useful under controlled conditions; it does not show how agents should learn such policies.

Fourth, the oracle is not a realistic architecture. It is included only to show how far the recursive agent remains from perfect repair.

Fifth, the viability functional is task-local. It should not be read as a universal measure of consciousness, intelligence or observerhood.

10 Reproducibility statement

The source package accompanying this publication contains a single-file Python implementation of the decision process, conditions, agents, fixed-seed experiment runner, summary generation and figure generation. It includes the three main paper figures together with additional generated viability, survival-rate and self-error figures used as consistency outputs. Running the default command

```
python mirror_lab_v.py --episodes 1400 --seed 780000 --outdir outputs/lab_v
```

reproduces the 39,200-episode dataset, summary table, recursive advantage table and figures used in this paper.

11 Conclusion

Recursive reliability is conditionally useful. It improves viability when first-order reliability is corrupted and meta-diagnosis is affordable. It becomes harmful when first-order reliability is already truthful or when meta-diagnostic cost dominates avoided loss. Lab V therefore closes the pre-Mathematics-II reliability sequence: observer-relevant reliability is not merely self-estimation, not merely channel decomposition, not merely repair and not merely recursion. It is reliability that improves viability after cost.

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